

FeBT: A Feature Balancing Transformer for Corporate ESG Forecasting

Yawen Li , Mengyu Zhuang , Guanhua Ye , Yan Li, Junheng Wang , Jinyi Zhou, and Pengfei Zhang 

Abstract—Environmental, social, and governance (ESG) serves as a crucial indicator for evaluating firms in terms of sustainable development. However, the existing ESG evaluation systems suffer from limitations, such as narrow coverage, subjective bias, and lack of timeliness. Therefore, there is a pressing need to leverage machine learning methods to predict the ESG performance of firms using their publicly available data. Traditional machine learning models encounter the feature imbalance problem due to the heterogeneity in ESG-related features. Common approaches typically involve unfolding all features, thereby granting high-dimensional folding features greater exposure and accessibility to downstream models, which results in the neglect of low-dimensional features. To fill the research gap regarding fully using the heterogeneous features of enterprises to enhance AI-based ESG prediction performance, we propose the Feature Balancing Transformer (FeBT), a model based on autoencoders and Transformer blocks. FeBT incorporates a novel feature balancing technique that compresses and enhances high-dimensional features from imbalanced data into low-dimensional representations, thereby ensuring a more balanced impact of high-dimensional and low-dimensional features on the model's performance in the downstream ESG forecasting module. Extensive experiments verified the superior performance of FeBT compared with state-of-the-art methods in real-world ESG-related datasets and evidenced that our feature balancing module provides significant insights from high-dimensional folding features.

Index Terms—ESG forecasting, data masking, time series forecasting.

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I. INTRODUCTION

ENVIRONMENTAL, social, and governance (ESG) information is becoming increasingly prominent in financial markets. It places a significant emphasis on environmental preservation, social responsibility, and corporate governance, regarding them as crucial considerations in investment decisions [1]. Investors, both institutional and individual, now rely on ESG data to make more informed choices that align with their long-term sustainability goals. Firms are adopting ESG principles to guide their practices and attract socially responsible investors. Bloomberg's latest report indicates that more than 53 trillion assets under management will be flowing into ESG-labeled investment products by 2025. Consequently, there is a growing research interest in capturing ESG information and evaluating firms' ESG performance.

However, the existing ESG performance evaluation system has several unsolved issues. The primary issue is the lack of consistency in ESG ratings across various providers. Even reputable ESG rating agencies, such as Bloomberg, MSCI, and Thomson Reuters, may use different metrics when assessing a firm's ESG performance within the same criteria, which leads to rating disagreement [2], [3], [4]. The rating scored by experts also results in a high level of subjectivity. An ESG assessment system should take a wider range of objective information into account, including factors available in public datasets [5], [6]. A secondary issue is related to the time lag in ESG rating [7], which makes it difficult for stakeholders to acquire real-time or foreseeing insights, thereby hindering their informed decision-making based on ESG information. Moreover, most small and medium-sized enterprises do not voluntarily participate in ESG evaluation, which results in a delay in ESG information disclosure [8], [9]. Thus, it is necessary to create a predictive model that can evaluate a firm's ESG performance in real-time while also ensuring ease of replication using standardized metrics derived from extensive data sources.

The rapid advancement of artificial intelligence has demonstrated its profound impact on the field of enterprise metric prediction [10]. In previous studies, researchers have used diverse machine learning models, including decision trees, K -nearest neighbors (KNNs), and support vector machines (SVMs), to forecast corporate-related metrics [11], [12], [13], [14], [15], [16]. However, these approaches frequently struggle to effectively capture intricate shifts in firm characteristics [12]. For instance, the fundamental concept behind using a KNN for predicting corporate financial crises involves assigning individuals to clusters that encompass the majority of their "nearest neighbors"

within the dataset [17]. However, this approach has limitations when handling high-dimensional data or imbalanced samples. As the dimensionality of firm features increases, the distance between data points becomes more dispersed. Additionally, an imbalanced distribution of samples from different categories in a dataset can impact the stability of the KNN. Moreover, when the number of samples in a category is small, its representation in the neighborhood may diminish, thereby affecting prediction accuracy. In other investigations, researchers have sought to use an SVM for predicting corporate credit ratings [18], [19]. An SVM maps input vectors to a high-dimensional feature space, which transforms a complex problem into a simpler problem that can be addressed using a linear discriminant function. Although SVMs outperform multiple discriminant analysis, case-based reasoning, and three-layer fully connected backpropagation neural networks, in terms of performance [20], they require a substantial number of training samples and more intricate feature engineering. Consequently, when ESG data are limited or a multitude of noisy and redundant features exist, SVMs may encounter certain challenges. Additionally, researchers have proposed various heterogeneous integration models [5], [21], [22], [23], [24] to enhance prediction accuracy by combining the strengths of various models (e.g., XGBoost and LightGBM) for handling diverse features and refining the feature selection process. Despite this, the limited availability of training data that results from the delayed inception and incomplete disclosure of ESG scores may induce significant disparities between base models, thereby impacting their generalization capability.

Considering the general characteristics of firms, which encompass ESG scores within the same framework, these elements are statistically comparable and represent time-series data of the same caliber. In this study, we employ the forecasting model with Transformer blocks specifically for multivariate time-series forecasting. The Transformer blocks are based on a multi-head attention mechanism, which was initially proposed for natural language translation tasks [25]. It concurrently represents each element of the input sequence by considering its context (both future and past), while multiple attention heads focus on various representation subspaces, thereby capturing multiple aspects of relevance between input elements. This characteristic makes the Transformer well-suited to time series data; for example, it can capture multiple periodicities within a signal [26]. However, a major challenge in applying Transformer blocks is the considerable variation in dimensionalities among features that influence ESG scores. Integrating features of varying dimensions directly leads to an expanded attention matrix. This expansion forces the Transformer model to distribute its attention across a broader range of dimensions, which could potentially reduce its predictive efficiency.

To this end, we propose a novel model, the Feature Balancing Transformer (FeBT), specifically designed for corporate ESG forecasting. FeBT merges the strengths of the Transformer with a series of autoencoders, effectively circumventing the limitations inherent in existing models. The architecture of FeBT is composed of two integral components: the upstream Feature Balancing Module (FBM) and the downstream ESG Forecasting Module (EFM). Within the FBM, autoencoders are employed

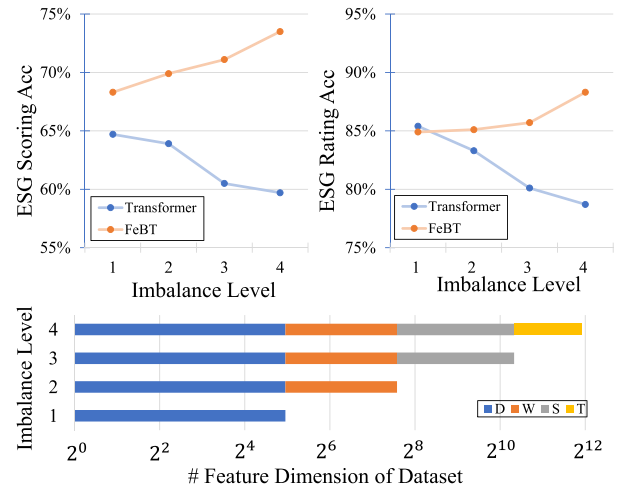


Fig. 1. Four datasets have been constructed at four distinct levels of imbalance, each employing a different combination of hybrid strategies. Level 1 encompasses solely Discrete Values (D), whereas Level 4 embodies a comprehensive blend of Discrete Values (D), Word Vectors (W), Sentiment Embeddings (S), and Time Series (T). The FeBT model mitigates the challenges posed by feature imbalance in tasks related to both ESG scoring and rating forecasting. Conversely, the conventional Transformer model experiences a drop in performance as the level of imbalance increases, despite being supplied with richer information.

to condense high-dimensional features—referred to as folding features in this study—thereby balancing the exposure of overall features. The EFM, on the other hand, leverages Transformer blocks to capture the long-range dependencies in the input data, culminating in accurate ESG score predictions. Furthermore, FeBT broadens the data horizon by utilizing a sliding window approach, incorporating historical insights from multidimensional feature data to refine and update the model parameters. As shown in Fig. 1, this approach significantly enhances the accuracy of predictions. The primary contributions of this study can be summarized as follows:

- We identify the issue of feature imbalance that may arise in ESG score prediction. Through an in-depth analysis of the feature distribution in the ESG-related task, we identify a significant imbalance between folding features (e.g., word vectors and time series variables) and plain features (e.g., discrete variables). Conventional Transformer models, which rely on attention mechanisms to weigh feature importance, may inadvertently focus more on the dense, high-dimensional folding features due to their larger volume of data points. As a result, plain features, which often contain critical but sparse information, can be under-emphasized during the model's learning process.
- We propose the FeBT model to address the challenges posed by limited data, numerous missing values, and dimensional inconsistency in the ESG prediction task. This model effectively uses the historical information of ESG scores to enhance prediction accuracy. To the best of our knowledge, this is the first machine learning model that comprehensively resolves the input data imbalance issue in the ESG prediction task, thereby facilitating transparent, replicable, and efficient ESG scores and predictions.

- We carried out a comprehensive set of experiments to determine the structure of the FeBT model and evaluate its effectiveness. The experimental results indicate that our FeBT model effectively counteracts feature imbalance. Impressively, it outperforms current state-of-the-art methodologies in delivering superior accuracy in ESG score predictions.

II. RELATED WORK

In this section, we analyze and summarize previous research that is relevant to our study.

A. ESG Forecasting

In recent years, there has been significant research activity in the field of ESG prediction for firms, with scholars endeavoring to enhance the feasibility and accuracy of this prediction through data and methodological advancements. Multiple studies have provided evidence that non-ESG information, including financial data and general characteristics, can serve as effective predictors of firms' ESG performance. For instance, notable correlations have been observed between features such as balance sheet information [27], annual corporate earnings [28], stock returns [29], and ESG scores, thereby highlighting their positive association. Additionally, in studies, researchers have used content analysis and text mining techniques to acquire social responsibility reports and social media data, considering sources such as firms' general characteristics, to predict ESG outcomes [28], [30], [31]. Despite identifying a range of meaningful predictors for ESG scores and attaining favorable predictive outcomes within specific evaluation systems, in these studies, the researchers overlooked the temporal dynamics of time-series data and the evolutionary patterns inherent in ESG scores. Consequently, the applicability and generalizability of the identified features and adopted models may be limited when applied to diverse ESG evaluation systems.

Another area of research centers on forecasting firms' financial or innovation performance using ESG data through various machine learning models, such as linear regression [32], random forest [27], support vector machine [33], and XGBoost [34]. Li et al. [34] found that, in terms of accuracy and interpretability, XGBoost outperformed alternative models when comparing their prediction results for innovation performance. Despite the input and output reversal in comparison with our study, we use their results as one of the baselines for our experiments because of the similarity between our experimental setups and theirs.

In the ESG prediction task, in certain studies, researchers have suggested enhancing the stacking model's base learner through weighted summation using accuracy as a basis during cross-validation. Furthermore, they have incorporated LightGBM feature selection into the training framework of the meta-learner's second layer to leverage the inherent structural characteristics of the ESG score [21]. Although this model attains higher accuracy through feature selection compared with single model predictions, it remains unable to address the issue of metric subjectivity in the evaluation. Conversely, Krappel et al. [5] introduced a heterogeneous integration model that comprises a feedforward

neural network, CatBoost, and XGBoost ensemble members. This model enables the automatic generation of ESG scores using publicly available data without relying on ESG information. The heterogeneous integration model involves combining multiple member models and requires an ample amount of training data to adequately learn the parameters and weights of these models. Despite this, in markets with incomplete disclosure, the scarcity of firm data or missing features may impede the model's ability to generalize effectively to new samples.

B. Feature Imbalance Solutions

Feature imbalance, characterized by significant disparities in dimensions among different features, presents a notable challenge in data modeling. McElfresh et al. [35] found that the performance difference between NNs and GBDTs is minimal and depends more on dataset characteristics. Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA) are common dimensionality reduction techniques used to mitigate this issue. While PCA effectively reduces the feature space, it may overlook features that are less variant but still important, potentially leading to loss of valuable information [36]. Similarly, LDA is limited by its assumption of linear separability and normal distribution of classes [37]. Regularization techniques, such as L1 and L2 regularization, assist in balancing feature influence by penalizing large weights. However, L1 regularization can lead to a sparse solution, sometimes excluding relevant features, while L2 regularization may not be effective in cases of high multicollinearity [38]. Feature weighting assigns varying importance to features based on their relevance. This method requires careful tuning; otherwise, it risks amplifying noise and irrelevant features [39]. To address these limitations, we use an autoencoder-based feature balancing module to solve the feature imbalance issue.

C. Autoencoders

Autoencoders, recognized for their proficiency in data dimensionality reduction and clustering, use an encoder-decoder framework to effectively compress data into a latent space, surpassing traditional methods like PCA by exploiting non-linear data relationships [40]. These models excel in handling complex datasets and have evolved through the incorporation of regularization techniques, enhancing their data representation capabilities. Notable developments include GMVAE and VaDE, which have shown exceptional unsupervised clustering performance in image datasets [41], [42]. Furthermore, the integration with other neural architectures has led to more nuanced latent variable interpretations. The CAE, combining autoencoders with CNNs, excels in deep clustering of image data [43], while AAEs, merging GANs with VAEs, offer improved clustering performance [44]. The MAE's introduction of random sampling and the Transformer to autoencoders has further enhanced performance, particularly on the ImageNet-1K dataset [45]. Additionally, self-supervised learning methods, such as SwitchTab, have made significant strides in capturing latent dependencies within tabular data [46]. Building on these

advancements, our FeBT introduces masked autoencoders to perform feature compression.

D. Time Series Analysis Models

Early time series analysis efforts, such as the ARIMA model [47], faced limitations due to linear assumptions and poor scalability. Alternative methods like matrix decomposition [48] improved learning across time series but struggled with complex data correlations. Addressing these shortcomings, hierarchical Bayesian formulations [49] and innovations like DeepAR [50], which combines AR models with RNNs in an encoder-decoder architecture, enhanced multidimensional time series prediction. Wen et al.'s approach [51] further mitigated error accumulation in such models.

The advent of the Transformer revolutionized multivariate time series forecasting. Chen et al. [52] applied a Transformer-based model for anomaly prediction, achieving notable results on benchmark datasets. Jin et al. [53] introduced a generative-style decoder, accelerating the Transformer's training and inference processes. The Informer model by Zhou et al. [54] tackled long time-series prediction challenges with its ProbSparse attention mechanism and generative decoder, addressing limitations in traditional Transformer architectures. Similarly, Zerveas et al. [26] proposed a Transformer-based framework with upstream unsupervised pre-training, significantly boosting performance. In our work, we leverage the transformer blocks to conduct downstream precise predictions in ESG forecasting tasks.

III. METHODOLOGY

In this section, we clarify definitions related to ESG prediction and introduce each part of FeBT in detail.

A. Preliminaries

First, we introduce the concept of imbalanced features, the characteristics of the ESG dataset, and the definition of the problem.

Definition 1 (Plain and Folding Feature): For N entities under scrutiny for ESG performance, we construe data from each assessment period (typically annual) as a distinct sample that comprises both plain features and folding features. In our ESG dataset, plain features, such as annual profit, have only one data point for each entity per period. Conversely, folding features, such as the local daily air pollution index, encompass a multiplicity of data points for each entity within the same time frame. We define plain features as $\mathbf{X}^{\text{pln}} = \{\mathbf{P}_1, \mathbf{P}_2, \mathbf{P}_3, \dots\}$ and folding features as $\mathbf{X}^{\text{fld}} = \{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots\}$, where $\mathbf{P}_i = \mathbf{x}_{(0,i)} \in \mathbb{R}^N$ and $\mathbf{F}_i = \{\mathbf{x}_{(i,1)}, \mathbf{x}_{(i,2)}, \mathbf{x}_{(i,3)}, \dots\} (i \in \mathbb{N}^+)$. Notably, $\mathbf{x}_{(0,1)}$ and $\mathbf{x}_{(0,2)}$ represent two disparate attributes, whereas $\mathbf{x}_{(i,1)}$ and $\mathbf{x}_{(i,2)}$ correspond to the varied dimensions of an identical attribute.

Definition 2 (Imbalanced ESG Dataset): Consider the ESG dataset $E \in \mathbb{R}^{N \times T}$, which encompasses N entities across T periods. The features and label of all entities at period t can be denoted by $\{\mathbf{X}_t^{\text{pln}}, \mathbf{X}_t^{\text{fld}}\}$ and $\mathbf{y}_t \in \mathbb{R}^N$, respectively. $\mathbf{y}_t$ denotes the ESG scores assigned by a third-party evaluator. Within

ESG datasets, the cardinality inequality ($|\mathbf{F}_{(t,i)}| \gg |\mathbf{P}_{(t,i)}| = 1$) causes a disparity between plain and folding features. Plain features are disadvantaged by a solitary instance of recognition in the attention layers of the Transformer, whereas folding features have increased visibility. Moreover, the exhaustive unfolding of attributes potentially diminishes the intrinsic patterns within folding features. Thus, the original ESG feature ensemble, defined for period t as:

$$\chi_t^{\text{orig}} = \{\mathbf{P}_{(t,1)}, \mathbf{P}_{(t,2)}, \dots, \mathbf{F}_{(t,1)}, \mathbf{F}_{(t,2)}, \dots\}, \quad (1)$$

is deemed unbalanced.

Definition 3 (Feature Balancing): The feature balancing operation is denoted as $B(d, \cdot)$. For any set of folding features $\mathbf{F}$, this operation results in a compressed set:

$$\mathbf{F}' = B(d, \mathbf{F}), \quad (2)$$

where d is the dimensionality compression coefficient which adjusts the compression degree and $d = |\mathbf{F}'| \ll |\mathbf{F}|$. $\mathbf{F}'$ is required to retain essential information from $\mathbf{F}$ to support downstream ESG forecasting. Consequently, the balanced dataset χ_t^{bal} supersedes the original χ_t^{orig} as the input for subsequent analytical tasks:

$$\chi_t^{\text{bal}} = \{\mathbf{P}_{(t,1)}, \mathbf{P}_{(t,2)}, \dots, \mathbf{F}'_{(t,1)}, \mathbf{F}'_{(t,2)}, \dots\}. \quad (3)$$

Although the folding features are processed independently by the autoencoder during pre-training, the resulting balanced dataset is then fed into the prediction module in a combined manner. This transformation plays a crucial role in enabling the attention mechanism to capture the interrelationships among features, particularly those with temporal dependencies. The feature balancing operation reduces the dimensionality of the input, helping streamline the model while ensuring that key temporal patterns and inter-feature correlations are preserved. This is especially important for time-series data, where the relationships between features evolve over time. By simplifying the feature space without losing critical information, the model enables the attention mechanism to effectively identify and process these complex dependencies, thereby improving performance in downstream forecasting tasks, such as ESG predictions.

Problem Definition: Corporate ESG Forecasting on an Imbalanced Feature Dataset. Given an imbalanced ESG dataset that integrates plain and folding features $\{\chi_t^{\text{orig}} | t = 1, 2, \dots, T\}$, our objective is to design a predictive model. This model incorporates a self-supervised feature balancing module (FBM) to execute $B(d, \cdot)$ and then forecast the future ESG outcomes $\mathbf{y}_{T+1}$. To simplify the expression, we omit the time subscript t to represent the aggregation of all periods (e.g., $\chi^{\text{orig}} = \{\chi_t^{\text{orig}} | t = 1, 2, \dots, T\}$) in following sections.

B. Overview of FeBT

The operational schema of FeBT is depicted in Fig. 2. The workflow of FeBT is divided into two sequential phases: an initial pre-training phase confined to the feature balancing module, followed by a forecasting phase that engages the entire model. The FBM is located on the lower left in the figure and

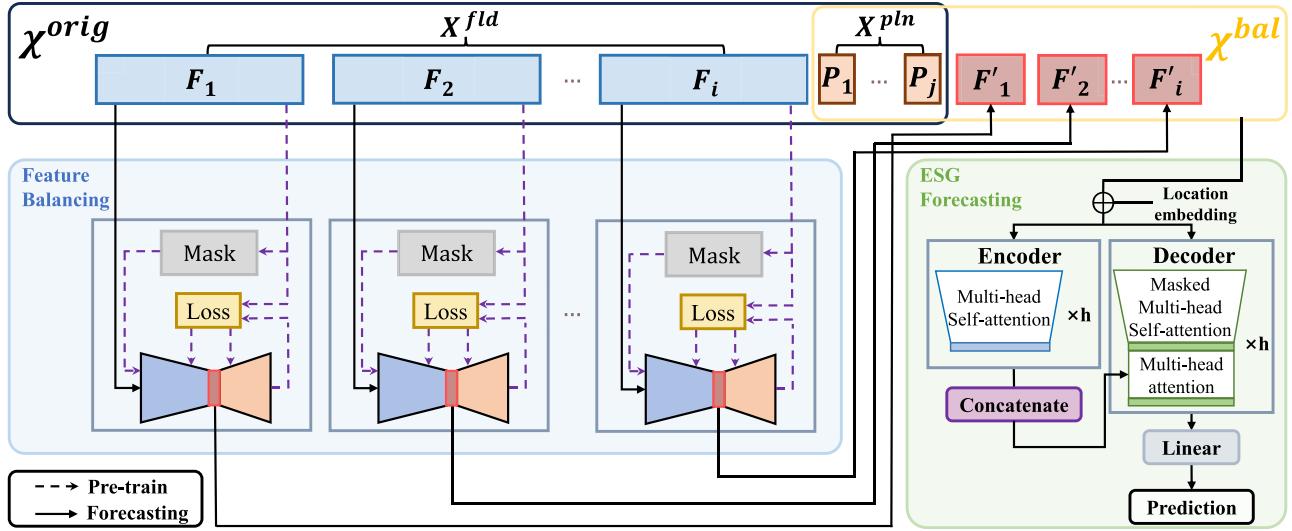


Fig. 2. Workflow of FeBT. The original dataset χ^{orig} contains folding features $\mathbf{X}^{fld}$ and plain features $\mathbf{X}^{pln}$. The upstream Feature Balancing Module generates latent representations F'_1 to F'_i for all folding features and restructures the balanced dataset χ^{bal} . The downstream ESG Forecasting Module then yields the prediction based on χ^{bal} .

takes the folding features as input. It undertakes the pretraining of $A = |\mathbf{X}^{fld}|$ distinct autoencoders for A folding attributes, thereby deducing an optimal compression strategy that corresponds to each specified d_i through a detached operation. Enhanced by data masking techniques, this module seeks to amplify the autoencoders' proficiency in pattern distillation. After pre-training, the downstream Transformer harnesses the balanced dataset χ^{bal} to extrapolate the final ESG predictions.

C. Pre-training

We assign each folding feature in our dataset to a specific autoencoder. This design allows each autoencoder to specialize in compressing a particular aspect of the raw data, thus ensuring more effective feature representation. The pre-training of these autoencoders is a crucial step before we start to train the predictive model (dashed line in Fig. 2). During this phase, we train autoencoders on their respective folding features, with the objective of learning to reconstruct these features from their compressed representations. This training process enables each autoencoder to develop a unique set of weights, which effectively captures the essential attributes of each feature, thereby facilitating a more nuanced and comprehensive understanding of the data.

Masking: To further enhance the quality of the compressed features, we use a masking technique shown in Fig. 3. This technique involves deliberately erasing parts of the input features during the training of the autoencoders. The erased parts, or masks, compel the autoencoders to focus on less obvious characteristics of the data, thereby improving their ability to reconstruct the original signal, even in the presence of incomplete or distorted input. This added complexity not only refines the feature representations but also leads to a degree

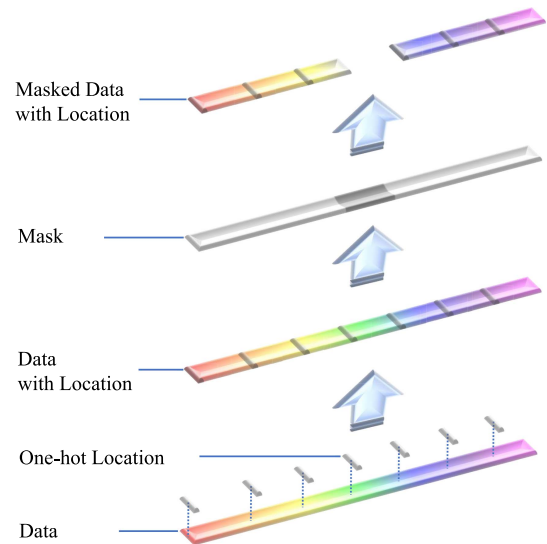


Fig. 3. Data masking procedure. The masked patch is invisible to the encoders in the Feature Balancing Module.

of robustness in the model. Specifically, we divide the folding feature vectors into k regular non-overlapping patches and make one patch of them invisible to the encoders. We adopt one-hot positional embeddings to identify the location of patches.

Autoencoder: Unlike typical encoders that handle the entire input data at one time, our encoder takes one patch of data as the input and repeats this behavior to generate k fragments of the latent representation. The decoder then produces the reconstruction result using the assembly of these fragments. For any autoencoder in FBM, it is compelled to determine the optimal parameter set for both the encoder and decoder to refine

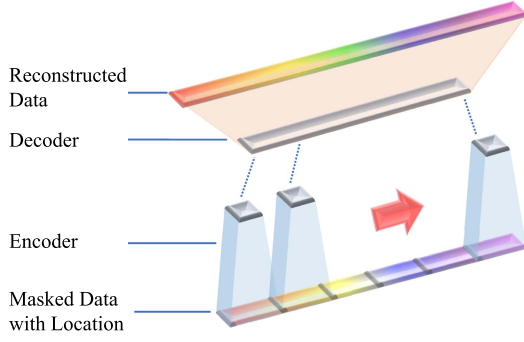


Fig. 4. Autoencoders in the pre-training phase. The input is masked data and the target is original data.

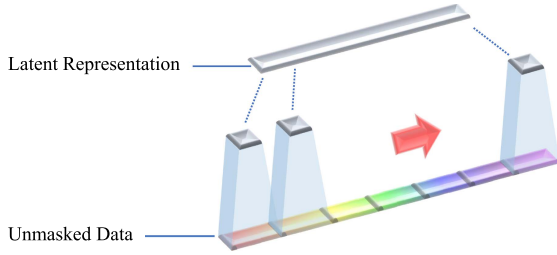


Fig. 5. Autoencoders in the forecasting phase. The decoder is removed and the parameters in the encoder are fixed. The input is the unmasked data and the output is the latent representation.

inter-attribute patterns from the raw signal by minimizing:

$$\mathcal{L}^* = \min_{\gamma, \theta} \mathcal{L}(f_{\gamma}(g_{\theta}(m(\mathbf{F}_i))), \mathbf{F}_i), \quad (4)$$

where $f_{\gamma}(\cdot)$, $g_{\theta}(\cdot)$ and $m(\cdot)$ are the decoder, encoder, and the masking operation, respectively. Fig. 4 illustrates the workflow of one autoencoder. Considering the limited data volume of individual folding features, we adopt CNN-based residual blocks in both the encoder and decoder to provide beneficial inductive bias. Note that we only use the decoder in the pre-training phase.

D. Forecasting

In this phase, FeBT restructures the ESG dataset with plain features and balanced folding features, and then yields the final prediction result.

Restructuring: As shown in Fig. 5, all the patches of the raw data pass through the encoder without any mask to generate the compressed latent representation. We denote the folding feature that has undergone FBM by $\mathbf{F}'_i$. After completing all these preparation steps, we can finally restructure the balanced dataset χ^{bal} with both plain features and refined folding features. χ^{bal} is handled by a predictive model based on Transformer blocks, which we describe next.

Deduction: We devise a streamlined predictive model that integrates a sequence of Transformer blocks, as delineated in the right half of Fig. 2. Specifically, we adopt the multi-head attention mechanism. Central to our design is the incorporation of the multi-head attention mechanism. The model initiates by projecting the input data, denoted by χ^{bal} , across multiple distinct linear spaces. These data are then segmented into h

different heads, where each head is transformed via distinct weight matrices W_i^Q , W_i^K , and W_i^V for $i = 1, 2, \dots, h$. Consequently, we obtain the sets $Q_i = \chi^{\text{bal}} \cdot W_i^Q$, $K_i = \chi^{\text{bal}} \cdot W_i^K$, and $V_i = \chi^{\text{bal}} \cdot W_i^V$, which represent the query, key, and value components, respectively, which are essential in the attention mechanism. Q_i encapsulates the vector of the current year's data, thereby facilitating the computation of similarity or correlation with data from all other years. K_i mirrors this process for all other years' data, correlating with Q_i , whereas V_i computes the weighted sum based on other years' data.

The scaled dot-product attention for each head is computed as follows:

$$H_i = \text{softmax} \left(\frac{Q_i K_i^T}{\sqrt{\eta}} \right) \cdot V_i, i = 1, 2, \dots, h \quad (5)$$

This vector is a new representation of the current year's data that can capture information from previous and subsequent years, where η is the dimension of Q_i and K_i . The results of the h heads are merged and then a linear projection is performed to obtain the final multi-head attention output:

$$M = H_1 \oplus H_2 \oplus \dots \oplus H_h \cdot W_O \quad (6)$$

where W_O is the output weight matrix. The multi-head attention mechanism can focus on different aspects of the data features, and its result M is passed through a linear layer to obtain the final predicted value $\hat{l}_{T+1}$. We use the mean squared error (MSE) as the loss function for the prediction task. During the training process, we update the model parameters through gradient descent to minimize the loss function. Thus, the model can learn the data distribution and thus obtain more accurate results in future predictions.

IV. EXPERIMENTS

In this section, we detail the data sources, evaluation metrics, and performance benchmarks used in our experiments, which are designed to address the following Research Questions (RQ):

- RQ1:** Can FeBT outperform SOTA models in ESG prediction?
- RQ2:** Did the FBM provide more insight than other feature balancing methods?
- RQ3:** How does the work efficiency of FeBT compare to that of traditional Transformer?
- RQ4:** How does feature imbalance hinder model performance?
- RQ5:** What is the contribution of each nontrivial technical detail in FeBT?

A. Datasets and Evaluation Metrics

We acquired four datasets through direct releases from companies or by gathering data via textual analysis. These datasets are the Chinese Major Cities Air Quality dataset (CMCAQ), the Stock Bar Information of Chinese Listed Companies (SBICIC), the Green Word Vectors of Chinese Listed Companies (GWV-CIC), and the Fundamental dataset of Chinese Listed Companies (FCIC). CMCAQ encompasses daily air quality data from 149 major cities in China. This dataset includes seven key indicators

TABLE I
FEATURES STATISTICS

Source	Feature Type	#Feature	#Dimension Per Annual	Fold
CMCAQ	<i>Time Series</i>	7	2,555	✓
SBICIC	<i>Sentiment Embedding</i>	3	1,095	✓
GWVCIC	<i>Word Vector</i>	1	160	✓
FCIC	<i>Discrete</i>	31	31	×

used to assess air quality, namely AQI (Air Quality Index), PM2.5, PM10, SO₂, NO₂, CO, and O₃. SBICIC contains information from the stock bar of Chinese-listed companies spanning from 2009 to 2021. It includes the daily volume of positive posts, the daily volume of negative posts, and the total daily readership of all posts. GWVCIC is derived from the textual analysis of the publicly disclosed information of listed companies. It extracts 160 word vectors related to green strategies and counts their occurrences in corporate public documents over the course of a year. This is used to reflect the extent of a company’s emphasis on green strategies. FCIC contains fundamental information about listed companies, which includes the characteristics of corporations (e.g., company type, number of employees, and R&D investment), CEO characteristics (including gender, age, and educational background), geographical features (e.g., regional GDP and growth rate, and intensity of local government’s green policies), and ESG characteristics (ESG scores and ratings, and scores and ratings for the ESG subcategory ‘E’). From the aforementioned four datasets, we compiled a total of 28,560 company records to use in the experiment. In Table I, we list some key statistics of features in all datasets.

For the ESG scoring regression task, we adopted Mean Squared Error (MSE) as the loss function, recognizing its effectiveness in capturing the variance of the prediction errors. We evaluated the regression performance using a set of carefully chosen metrics.

Scoring 5% Margin Acc: This metric quantifies the proportion of predictions where the predicted ESG scores deviate by no more than 5% from their groundtruth. It offers a fine-grained assessment of the model’s performance with a narrow error margin.

Scoring 10% Margin Acc: Similar to the 5% margin metric, this measures the proportion of predictions within a 10% error margin. It provides a broader view of the model’s accuracy, accommodating a slightly more relaxed error tolerance.

Scoring MAPE: This represents the mean of the absolute percentage errors across all predictions, offering a comprehensive overview of the model’s overall percentage accuracy.

In parallel, we extended our investigation to ESG rating classification to gain additional insights into our model’s versatility. Here, we employed cross-entropy as the loss function, acknowledging its suitability for classification problems. The classification performance was evaluated based on the accuracy (**Rating Acc**), macro average recall (**Rating Rec**), and macro average precision (**Rating Pre**).

B. Baselines and Model Configurations

In our assessment of effectiveness, we compared our FeBT model against five established baseline models:

- **XGBoost [55]:** XGBoost, a gradient boosting tree (GBT)-based machine learning algorithm, constructs a robust ensemble model by integrating multiple decision trees.
- **LightGBM [56]:** LightGBM, another GBT-based algorithm, distinguishes itself from XGBoost through its histogram-based and parallel optimization techniques.
- **LSTM [57]:** The Long Short-Term Memory (LSTM) model, advancing beyond traditional Recurrent Neural Networks (RNN) [58], incorporates memory cells and gating mechanisms.
- **GRU [59]:** The Gated Recurrent Unit (GRU) model, a streamlined variant of the RNN, modifies the LSTM architecture by consolidating its gating mechanisms.
- **Transformer:** The ESG forecasting module of the FeBT framework. It is trained on the original dataset and operates as a standalone Transformer model.

The baseline models were configured with default settings. The optimal hyperparameters of FeBT are determined via grid search, including dimensionality compression coefficient d , the number of attention heads of Transformer blocks h , the input sequence length of Transformer blocks l_{seq} . The search space and the optimal value of each hyperparameter is listed in Table III.

C. ESG Forecasting Effectiveness

Horizontal Comparison (RQ1): Based on the outcomes in Table II, the FeBT model has demonstrated exceptional performance across all metrics. During the regression task, an intriguing observation was made across all baseline models: while they exhibited commendable performance in the Scoring 10% Margin Accuracy, a significant decline in performance was noted when the margin was reduced to 5%. The average performance degradation of all baselines was quantified at 28.5%. In stark contrast, our FeBT model exhibited a much smaller performance differential of only 18.1% between these two experimental conditions. For the most challenging finer-grained regression metric, FeBT presents distinguished performance, outshining its nearest rival by 8.9%. This underscores FeBT’s potential in handling complex tasks. In the classification task for ESG rating, the performance gap between the FeBT model and baselines narrows. This phenomenon can be primarily attributed to the relatively less complex nature of classification tasks compared to regression tasks. An observation from Table II provides further insight: FeBT’s superior performance in metrics sensitive to error magnitude, such as MAPE, suggests that its design prioritizes minimizing large errors through a more conservative prediction strategy. This approach reduces the occurrence of extreme outliers but leads to fewer predictions falling within the 10% margin compared to simpler models like LightGBM. The latter models, which exhibit higher variance, tend to produce more predictions clustered around the true value, allowing more instances to meet the broader 10% margin. In contrast, FeBT’s focus on minimizing large deviations results in fewer predictions

TABLE II
ESG SCORE AND RATING PREDICTION PERFORMANCE COMPARISON

Model	Scoring 5% Margin Acc $\uparrow$	Scoring 10% Margin Acc $\uparrow$	Scoring MAPE $\downarrow$	Rating Acc $\uparrow$	Rating Rec $\uparrow$	Rating Pre $\uparrow$
XGBoost	0.624	0.899	0.053	0.857	0.552	0.774
LightGBM	0.646	0.920	0.048	0.857	0.558	0.767
LSTM	0.590	0.865	0.059	0.814	0.568	0.699
GRU	0.581	0.884	0.057	0.794	0.598	0.660
Transformer	0.597	0.873	0.057	0.787	0.581	0.732
FeBT	0.735	0.916	0.041	0.883	0.660	0.795

TABLE III
FeBT HYPERPARAMETER CONFIGURATION

Config	Search Space	Optimal
d	{8, 16, 24, 32, 64, 128, 256}	16
h	{1, 2, 3, 4, 5, 6, 7, 8}	4
l_{seq}	{2, 4, 6, 8, 10}	6

within the 10% margin, as it emphasizes precision at smaller margins, making its predictions more conservative and narrower.

Another observation was the underperformance of temporal baseline models (such as LSTM and GRU) in contrast to GBT-based models in both the ESG regression and classification contexts. This underperformance likely stems from the heightened sensitivity of temporal models to the feature imbalance present within the ESG dataset. Further comparative analysis, particularly between the Transformer model and our FeBT model, sheds more light on this issue. The integration of the Feature Balance Module (FBM) into the FeBT model has led to a significant uplift in performance metrics. Specifically, there was an improvement of 14.2% in Scoring 5% Margin Accuracy and an 8.8% increase in Rating Accuracy. These enhancements not only corroborate our hypothesis regarding the adverse impact of feature imbalance in ESG data but also demonstrate the effectiveness of the FBM in mitigating this challenge.

Feature Balancing Capability (RQ2): The ESG dataset used for prediction encompasses four distinct feature types: Discrete (D), Word Vector (W), Sentiment Embedding (S), and Time Series (T). To investigate the impact of feature imbalance in the dataset on ESG prediction task and to evaluate the feature-balancing capability of the FBM, we reconstructed four datasets, classifying their imbalance levels based on the included feature types: only D (1), both D and W (2), D, W, and S (3), and all feature categories (4). Under varying imbalance levels, we compared the performance of the Transformer and FeBT in terms of Scoring 5% Margin Accuracy, Scoring MAPE, and Rating Accuracy, with results presented in Table IV. It is evident that with increasing imbalance levels, FeBT demonstrates increasingly robust predictive capabilities, effectively addressing a challenge that traditional Transformers are unable to handle.

In an alternative approach, feature balancing was also attempted by either padding or utilizing Principal Component Analysis (PCA) to match the dimensionality of plain features with that of folding features. The performance of these methods, denoted as ‘Padding+Transformer’ and ‘PCA+Transformer’ in

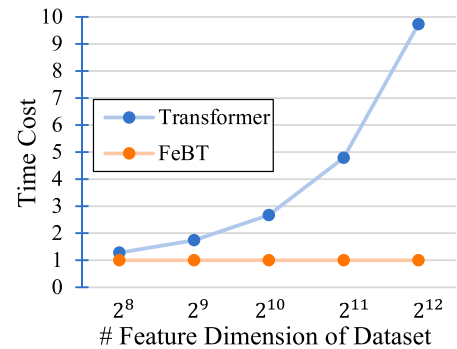


Fig. 6. Comparison of FeBT and Transformer in terms of work efficiency across various dataset sizes.

Table V, exhibited noticeable improvements over the Transformer baseline, particularly in Scoring MAPE and Rating Accuracy. However, it is crucial to highlight that there remains a substantial performance gap when compared to the standard FeBT model. This disparity underscores the fact that our FBM does more than just address feature imbalance; it also contributes valuable insights into the nature of the folding features, thereby enhancing the overall predictive capability of the model.

Work Efficiency (RQ3): We assess FeBT’s computational overhead and scalability relative to a traditional Transformer (Table VI). Both models employ attention mechanisms with a complexity of $O(L^2 \times d)$ —where L is the sequence length and d is the dimension—making them susceptible to bottlenecks with long sequences due to quadratic growth in computations.

FeBT addresses this challenge by reducing the effective sequence length from L to $\frac{L}{s^n}$ using its Feature-Balanced Module (FBM), where s is the convolution stride and n is the number of convolution layers. This reduction significantly lowers the computational load in the self-attention mechanism, as illustrated in Fig. 6 (experiments performed on Tesla V100-SXM2-32GB). In contrast to the traditional Transformer, whose time cost escalates sharply with increasing dataset size, FeBT maintains substantially lower overhead.

Thus, in terms of scalability and efficiency, FeBT offers a more practical solution for processing long sequences and large-scale data, outperforming the traditional Transformer in both performance and resource consumption.

TABLE IV
COMPARISON OF TRANSFORMER AND FEBT MODELS ACROSS IMBALANCE LEVELS

Imbalance Level*	Transformer			FeBT		
	Scoring 5% Margin Acc ↑	Scoring MAPE ↓	Rating Acc ↑	Scoring 5% Margin Acc ↑	Scoring MAPE ↓	Rating Acc ↑
1	0.647	0.047	0.854	0.683	0.054	0.849
2	0.639	0.049	0.833	0.699	0.049	0.851
3	0.605	0.055	0.801	0.711	0.043	0.857
4	0.597	0.057	0.787	0.735	0.041	0.883

* Imbalance levels are defined by the number of feature types in the dataset as follows: Level 1 (D), Level 2 (D+W), Level 3 (D+W+S), and Level 4 (D+W+S+T), where D = Discrete Values, W = Word Vectors, S = Sentiment Embeddings, and T = Time Series.

TABLE V
INFLUENCE OF FEATURE BALANCING METHODS

Feature Balancing Method	Scoring 5% Margin Acc ↑	Scoring MAPE ↓	Rating Accuracy ↑
Transformer	0.597	0.057	0.787
PCA+Transformer	+0.087	-0.003	+0.056
Padding+Transformer	+0.089	-0.004	+0.064
FeBT	+0.138	-0.016	+0.096

TABLE VI
TIME COMPLEXITY COMPARISON OF TRANSFORMER AND FEBT

Model Type	Time Complexity
Transformer	$O(L^2 \times d)$
FeBT	$O\left(\frac{L}{s} \times k + \left(\frac{L}{s^n}\right)^2 \times d\right)$

Note: L is the input sequence length, s is the convolution stride in the FBM, k is the kernel size, n is the number of convolution layers, and d is the representation dimension.

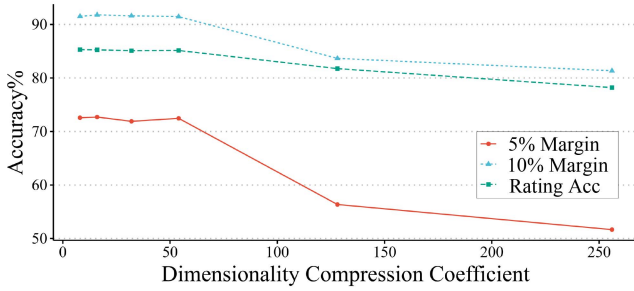


Fig. 7. Influence of dimensionality compression coefficient d in autoencoders.

D. Hyperparameter Sensitivity

Dimensionality Compression Coefficient (RQ4): The dimensionality compression coefficient, denoted as d , represents the dimension number of F' that compressed by the feature balancing module. To explore how this coefficient impacts the overall predictive performance, we experimented with varying values of d , specifically 8, 16, 24, 32, 64, 128, and 256. The effects of these variations on the scoring 5% margin accuracy, scoring 10% margin accuracy, and rating accuracy are presented in Fig. 7.

A key finding from this analysis is a consistent decline in performance when d increases from 16. This trend aligns with

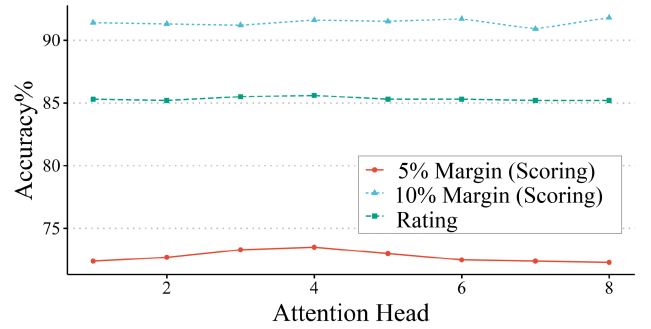


Fig. 8. Influence of attention head number h in Transformer blocks.

our initial hypothesis that Transformer blocks are optimized through the feature balancing process. Notably, the highest accuracy was achieved at $d = 16$, which is established as the default setting for the FeBT model. Furthermore, the performance of the FBM remains relatively stable for d values less than 32, underscoring the efficacy of autoencoders in efficiently refining and compressing the folding features.

Attention Head Number Analysis: Central to the FeBT framework's ESG forecasting module is the multi-head attention mechanism within the Transformer blocks, designed to learn diverse attention weight distributions. To evaluate the impact of varying the number of attention heads, denoted as h , on the model's predictive performance, a series of experiments were conducted. The outcomes of these experiments are illustrated in Fig. 8. Analysis of the results reveals a key insight: the FeBT model exhibits a notable robustness to changes in h . This is evidenced by the relatively stable predictive performance across different configurations of the attention head number, indicating that FeBT's effectiveness is not heavily contingent on this particular hyperparameter.

E. Ablation Study

Data Masking and Positional Encoding (RQ5): The FBM incorporates two pivotal techniques, namely data masking and positional encoding, to augment the autoencoder's capacity. Data masking, a key element in this process, involves the deliberate obscuration of parts of the input data to the encoder. This intentional complexity in the data reconstruction process compels the encoder to evolve a more sophisticated feature extraction

TABLE VII
INFLUENCE OF DATA MASKING AND POSITIONAL EMBEDDING

Data Masking	Positional Embedding	Scoring 5% Margin Acc $\uparrow$	Scoring MAPE $\downarrow$	Rating Accuracy $\uparrow$
×	×	0.720	0.043	0.849
×	✓	+0.002	−0.001	+0.001
✓	×	+0.004	−0.001	+0.010
✓	✓	+0.015	−0.002	+0.034

mechanism. The result is a more robust latent representation, enriched with deeper semantic information. Positional encoding, on the other hand, is employed to articulate the sequential relationships inherent in the data. This is instrumental in refining the model's comprehension of the input sequence's structure.

To quantitatively assess the impact of data masking and positional encoding on compressing upstream continuous datasets, we conducted four distinct experimental sets. These experiments were guided by optimal structural parameters gleaned from preliminary investigations. The findings, as encapsulated in Table VII, demonstrate that each technique contributes positively to the module's performance Individually. When data masking is not applied, the inherent sequential relationships in the data remain more pronounced, allowing the model to effectively capture relevant temporal information from the context. This reduces the model's reliance on explicit positional information, resulting in a limited impact of positional encoding on accuracy [60]. However, when data masking is introduced, the sequential information in the data is partially obscured, making it more challenging for the model to interpret the order of elements. In this scenario, positional encoding plays a crucial role in preserving the model's ability to understand the sequence order, thereby enhancing its representational capacity. Furthermore, a clear synergistic effect between data masking and positional encoding within the FBM. This synergy underscores the integral role of these techniques in optimizing the model's efficiency and predictive prowess.

Tuning (RQ5): In the development of the FeBT model, we did not adopt fine-tuning for upstream FBM. Our work focused on the efficacy of the universal feature balancing capability, while the performance improvement from fine-tuning may compensate for the contrast. Alternatively, we integrated two fundamental yet efficient representation learning techniques — the linear probe [45] and the adapter [61] — to enhance the interoperability between the upstream FBM and the downstream EFM. The linear probe strategy (lin) involves freezing the parameters within the FBM while concurrently training it with the downstream Transformer blocks. In contrast, the adapter technique (adp) seeks to augment FeBT's performance by appending a specialized adapter block to each autoencoder within the FBM. These adapter blocks undergo training subsequent to the initial learning processes of both the FBM and EFM.

We examined the influence of linear probe (lin), adapter block (adp), and the hybrid, the results are detailed in Table VIII. Note that in the hybrid method, the linear probe is positioned beneath the adapter blocks. When compared to a direct connection approach devoid of any fine-tuning, the incorporation of the

TABLE VIII
INFLUENCE OF TUNNING METHODS

Tuning Method	Scoring 5% Margin Acc $\uparrow$	Scoring MAPE $\downarrow$	Rating Accuracy $\uparrow$
direct	0.639	0.048	0.837
lin	+0.036	−0.004	+0.000
adp	+0.041	−0.004	+0.002
hybrid	+0.096	−0.007	+0.046

linear probe and adapter techniques manifested improvements in Scoring 5% Margin Accuracy, Scoring MAPE, and Rating Accuracy by 9.6%, 0.007, and 4.6% respectively. Consequently, the optimal configuration for our FeBT model was established as a composite of both linear probe and adapter blocks, thereby optimizing the model's predictive accuracy and efficiency.

V. LIMITATIONS

Although FeBT has significantly advanced ESG forecasting by mitigating feature imbalance, several challenges remain. The model's heavy reliance on high-quality, comprehensive corporate data means that any data gaps or inaccuracies can compromise its performance. Moreover, the integration of autoencoders with Transformer blocks introduces substantial computational overhead. Additionally, optimizing the dimensionality compression coefficient in the Feature Balancing Module is challenging; suboptimal tuning may lead to the loss of important information. Recognizing the current limitations, our future work will focus on enhancing data preprocessing to better manage incomplete datasets, optimizing the model architecture to reduce computational demands, and developing more robust methods for tuning the Feature Balancing Module. These targeted improvements will not only elevate the practical performance of FeBT but also refine the theoretical framework underlying data-adaptive ESG forecasting.

VI. CONCLUSION

FeBT presents a novel, two-stage, data-adaptive framework that effectively improves corporate ESG predictions through sophisticated feature integration and imbalance mitigation. The model makes a theoretical contribution by proposing a universal feature balancing approach that deepens our understanding of integrating high-dimensional, heterogeneous data in predictive modeling. By combining autoencoders with Transformer blocks, FeBT offers fresh insights into the synergy between representation learning and sequence modeling in the context of imbalanced datasets.

VI. CONFLICT OF INTEREST

The authors declare no potential conflict of interests.

VI. DATA AVAILABILITY STATEMENT

The model was evaluated using publicly available benchmarks that can be accessed via their references.

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